

14/11/2025

$$\bullet \lim_{x \rightarrow +\infty} \frac{(1+x^2)^6 - x^{12}}{x^{10}} = \lim_{x \rightarrow +\infty} \frac{x^{12} \left[\left(1 + \frac{1}{x^2}\right)^6 - 1 \right]}{x^{12} \cdot \frac{1}{x^2}} = 6$$

$\rightarrow \lim_{y \rightarrow 0} \frac{(1+y)^6 - 1}{y}$

ALCUNI CONFRONTI FRA FUNZIONI ELEMENTARI:

TEO: SIA $a > 1$, $\alpha \in \mathbb{R}$. ALLORA:

$$\lim_{x \rightarrow +\infty} \frac{x^\alpha}{a^x} = 0$$

POTENZA È MENO POTENTE
ESPONENZIALE È PIÙ POTENTE

ES: $\lim_{x \rightarrow +\infty} \frac{x^{999999}}{(1,0000001)^x} = 0$

ALCUNE CONSEGUENZE: SIA $a > 1$, $\alpha > 0$

1) $\lim_{x \rightarrow -\infty} |x|^\alpha \cdot a^x \Rightarrow y = -x \Rightarrow \lim_{y \rightarrow +\infty} |y|^\alpha \cdot a^{-y} = \lim_{y \rightarrow +\infty} \frac{|y|^\alpha}{a^y} = 0$

2) $\beta > 0$, $\lim_{x \rightarrow +\infty} \frac{|\log_a x|^\beta}{x^\alpha} \Rightarrow y = \log_a x \Rightarrow \lim_{y \rightarrow +\infty} \frac{|y|^\beta}{a^{y\alpha}} = 0$
 $x = a^y$

3) $\beta > 0$, $\lim_{x \rightarrow 0^+} |\log_a x|^\beta x^\alpha \Rightarrow y = \log_a x \Rightarrow \lim_{y \rightarrow -\infty} |y|^\beta \cdot a^{y\alpha} = \lim_{y \rightarrow -\infty} \frac{|y|^\beta}{a^{-y\alpha}} = 0$
 $x = a^y$

ES: $\lim_{x \rightarrow +\infty} \frac{x^{10} + \log(x)}{e^x + x \log(x)} = \lim_{x \rightarrow +\infty} \frac{x^{10} \left(1 + \frac{\log(x)}{x^{10}}\right)}{e^x \left(1 + \frac{x \log(x)}{e^x}\right)} = \lim_{x \rightarrow +\infty} \frac{x^{10}}{e^x} = 0$

CONFRONTI ASINTOTICI:

DEF: SIA $A \subseteq \mathbb{R}$, $f, g: A \rightarrow \mathbb{R}$, $x_0 \in \mathbb{R}$ PUNTO DI ACC. PER A . SIANO $f, g \neq 0$ DEF. PER $x \rightarrow x_0$.

SI DICE CHE f È ASINTOTICA A g PER $x \rightarrow x_0$ SE:

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 1 \rightarrow \text{SONO SIMILI PER } x \rightarrow x_0$$

IN TAL CASO $f \sim g$ PER $x \rightarrow x_0$.

PROP: (SOSTITUZIONE FZ. ASINTOTICHE): [QUOZIENTI E PRODOTTI]

SUPPONIAMO CHE $f_1 \sim f_2$ E $g_1 \sim g_2$ PER $x \rightarrow x_0$

$$1) \text{ SE } \exists \lim_{x \rightarrow x_0} f_2 \cdot g_2 = l \Rightarrow \exists \lim_{x \rightarrow x_0} f_1 \cdot g_1 = l$$

$$2) \text{ SE } \exists \lim_{x \rightarrow x_0} \frac{f_2}{g_2} = l \Rightarrow \exists \lim_{x \rightarrow x_0} \frac{f_1}{g_1} = l$$

RELAZIONI ASINTOTICHE DAI LIMITI FONDAMENTALI:

$$1) \sin(x) \sim x \text{ PER } x \rightarrow 0 \quad \text{INFATTI} \quad \lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

$$2) 1 - \cos(x) \sim \frac{x^2}{2} \text{ PER } x \rightarrow 0 \quad \text{INFATTI} \quad \lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x^2} \cdot 2 = \frac{1}{2} \cdot 2 = 1$$

$$3) \log(1+x) \sim x \text{ PER } x \rightarrow 0$$

$$4) e^x - 1 \sim x \text{ PER } x \rightarrow 0$$

$$5) (1+x)^\alpha - 1 \sim \alpha x \text{ PER } x \rightarrow 0 \quad \text{INFATTI} \quad \lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} \cdot \frac{1}{\alpha} = \frac{\alpha}{\alpha} = 1$$

$$6) \tan(x) \sim x \text{ PER } x \rightarrow 0$$

$$7) \sinh(x) \sim x \text{ PER } x \rightarrow 0$$

$$8) \cosh(x) - 1 \sim \frac{x^2}{2} \text{ PER } x \rightarrow 0$$

$$9) \operatorname{arcsin}(x) \sim x \text{ PER } x \rightarrow 0$$

$$10) \operatorname{arctan}(x) \sim x \text{ PER } x \rightarrow 0$$

OSS: SE $f(x) \rightarrow 0$ PER $x \rightarrow x_0$, PER COMPOSIZIONE POSSIAMO OTTENERE:

$$\begin{aligned} \sin(f(x)) &\sim f(x) \quad x \rightarrow x_0 \\ 1 - \cos(f(x)) &\sim \frac{f^2(x)}{2} \quad x \rightarrow x_0 \end{aligned}$$

ES: $\lim_{x \rightarrow 0} \frac{\sin^2(2x)}{\tan(3x^2)} \Rightarrow$ LA RISOLVIAMO IN 2 MODI:

$$\begin{aligned} 1) \lim_{x \rightarrow 0} \frac{\sin^2(2x)}{\tan(3x^2)} &= \lim_{x \rightarrow 0} \frac{\overset{1^2=1}{\sin^2(2x)}}{(2x)^2} \cdot (2x)^2 \cdot \frac{1}{\tan(3x^2)} = \lim_{x \rightarrow 0} \frac{4x^2}{\tan(3x^2)} \cdot \frac{3}{4} \cdot \frac{4}{3} = \\ &= \lim_{x \rightarrow 0} \underbrace{\frac{3x^2}{\tan(3x^2)}}_1 \cdot \frac{4}{3} = \frac{4}{3} \end{aligned}$$

$$\begin{aligned} 2) \sin(x) &\sim x \quad \text{PER } x \rightarrow 0 \\ \tan(x) &\sim x \quad \text{PER } x \rightarrow 0 \end{aligned} \quad \begin{aligned} \sin^2(2x) &\sim (2x)^2 \quad \text{PER } x \rightarrow 0 \\ \tan(3x^2) &\sim 3x^2 \quad \text{PER } x \rightarrow 0 \end{aligned} \quad \lim_{x \rightarrow 0} \frac{(2x)^2}{3x^2} = \frac{4}{3}$$

FORME INDETERMINATE DI TIPO ESPONENZIALE: "1[∞]", "0⁰", "∞⁰"

SIANO $f, g: \mathbb{R} \rightarrow \mathbb{R}$, $f > 0$, $x_0 \in \mathbb{R}$. CONSIDERIAMO:

$$\lim_{x \rightarrow x_0} (f(x))^{g(x)}$$

$$f(x)^{g(x)} = e^{\log(f(x))^{g(x)}} = e^{g(x) \cdot \log(f(x))}$$

N.B.: NON È PIÙ UNA FORMA INDET. DI TIPO ESPONENZIALE MA DI TIPO PRODOTTO:

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{\sqrt{x}} + \frac{1}{x^2}\right)^{\sqrt[3]{x}} = \lim_{x \rightarrow +\infty} e^{\sqrt[3]{x} \cdot \underbrace{\log\left(1 + \frac{1}{\sqrt{x}} + \frac{1}{x^2}\right)}_{\rightarrow 0}} \Rightarrow " \infty \cdot 0 "$$

$$\hookrightarrow \lim_{x \rightarrow +\infty} \sqrt[3]{x} \cdot \frac{\log\left(1 + \frac{1}{\sqrt{x}} + \frac{1}{x^2}\right)}{\frac{1}{\sqrt{x}} + \frac{1}{x^2}} \cdot \frac{1}{\sqrt{x}} + \frac{1}{x^2} = \lim_{x \rightarrow +\infty} \sqrt[3]{x} \cdot \left(\frac{1}{\sqrt{x}} + \frac{1}{x^2}\right) =$$

$$= \lim_{x \rightarrow +\infty} x^{\frac{1}{3}} \left(x^{-\frac{1}{2}} + x^{-2}\right) = \lim_{x \rightarrow +\infty} \frac{1}{x^{1/6}} + \frac{1}{x^{5/3}} = 0$$

$$\lim_{x \rightarrow +\infty} e^{\sqrt[3]{x} \cdot \log\left(1 + \frac{1}{\sqrt{x}} + \frac{1}{x^2}\right)} = 1$$

ES 2: • $\lim_{x \rightarrow 0^+} x^{\sin(x)} \Rightarrow "0^0" \Rightarrow \lim_{x \rightarrow 0^+} e^{\sin(x) \cdot \log x} \Rightarrow " \infty \cdot 0 "$

$$\lim_{x \rightarrow 0^+} \sin(x) \cdot \log(x) = \lim_{x \rightarrow 0^+} \underbrace{\frac{\sin(x)}{x}}_1 \cdot \underbrace{x \cdot \log(x)}_0 = 0$$

POT. • LOG $\rightarrow y = \log(x) \rightarrow \log \frac{y}{e^y} = 0$
 $x = e^y$

$$\lim_{x \rightarrow 0^+} e^{\sin(x) \cdot \log(x)} = e^0 = 1$$

ES 3:

$$\lim_{x \rightarrow +\infty} (2^x + 1)^{\frac{3}{x}} " \infty^0 " \Rightarrow \lim_{x \rightarrow +\infty} e^{\frac{3}{x} \log(2^x + 1)} \Rightarrow " \infty \cdot 0 "$$

$$\lim_{x \rightarrow +\infty} \frac{3}{x} \cdot \log(2^x + 1) = \lim_{x \rightarrow +\infty} \frac{3}{x} \cdot \log\left(2^x \left(1 + \frac{1}{2^x}\right)\right) = \lim_{x \rightarrow +\infty} \frac{3}{x} (\log 2^x + \log(1 + \frac{1}{2^x})) =$$

$$= \lim_{x \rightarrow +\infty} \frac{3}{x} \left(x \log 2 + \log(1 + \frac{1}{2^x})\right) = \lim_{x \rightarrow +\infty} 3 \log 2 + \frac{3}{x} \cdot \frac{\log(1 + \frac{1}{2^x})}{\frac{1}{2^x}} \cdot \frac{1}{2^x} = \lim_{x \rightarrow +\infty} \frac{3}{x} \cdot \frac{1}{2^x} + 3 \log 2$$

$$= 3 \log 2 \Rightarrow \lim_{x \rightarrow +\infty} e^{\frac{3}{x} \log(2^x + 1)} = e^{3 \log 2} = e^{\log 2^3} = 2^3 = 8$$

ES: CONFRONTI ASINTOTICI:

$$\bullet \lim_{x \rightarrow 0} \frac{\lim_{x \rightarrow 0} (3x^3) \cdot \log(1+2x^4)}{\sinh(2x^3) [1-\cos(3x)]^2} \rightarrow \frac{0}{0}$$

$$\begin{array}{l} \lim_{x \rightarrow 0} (3x^3) \sim 3x^3 \quad x \rightarrow 0 \quad \left| \quad \log(1+2x^4) \sim 2x^4 \quad x \rightarrow 0 \right| \\ \sinh(2x^3) \sim 2x^3 \quad x \rightarrow 0 \quad \left| \quad (1-\cos(3x))^2 \sim \left[\frac{(3x)^2}{2}\right]^2 = \frac{81x^4}{4} \quad x \rightarrow 0 \right| \end{array}$$

$$\lim_{x \rightarrow 0} \frac{\cancel{3x^3} \cdot \cancel{2x^4}}{\cancel{2x^3} \cdot \frac{81\cancel{x^4}}{4}} = \frac{4}{27}$$

$$\bullet \lim_{x \rightarrow +\infty} \frac{4e^{2x} + \left(\frac{1}{2}\right)^x + 2\log_2 x - 3x^2}{3^{-2x} + x^{20} - 10^x} = \lim_{x \rightarrow +\infty} \frac{e^{2x} \left(4 + \frac{1}{2^x \cdot e^{2x}} + \frac{2\log_2 x}{e^{2x}} - \frac{3x^2}{e^{2x}}\right)}{10^x \left(\frac{1}{3^{2x} \cdot 10^x} + \frac{x^{20}}{10^x} - 1\right)} =$$

$$= \lim_{x \rightarrow +\infty} \frac{-4e^{2x}}{10^x} = \lim_{x \rightarrow +\infty} -4 \left(\frac{e^2}{10}\right)^x = 0$$

$$\hookrightarrow 0 < \frac{e^2}{10} < 1 \quad \hookrightarrow a$$

$$a^x \rightarrow 0 \quad x \rightarrow +\infty$$

$$\bullet \lim_{x \rightarrow 0^+} \frac{2^x - 3^x}{\log(2^x + x^2 \sin(\frac{1}{x}))} \Rightarrow \frac{0}{0} \Rightarrow \lim_{x \rightarrow 0^+} \frac{2^x - 1 + 1 - (3^x - 1 + 1)}{\log(2^x + x^2 \sin(\frac{1}{x}))} =$$

$$= \lim_{x \rightarrow 0^+} \frac{\left(\frac{2^x-1}{x}\right) \cdot x - \left(\frac{3^x-1}{x}\right) \cdot x}{\log(2^x + x^2 \sin(\frac{1}{x}))} = \lim_{x \rightarrow 0^+} \frac{x \log(2) - x \log(3)}{\log(2^x + x^2 \sin(\frac{1}{x}))}$$

$$= \lim_{x \rightarrow 0^+} \frac{x \log(2) - x \log(3)}{\log(2^x - 1 + 1 + x^2 \sin(\frac{1}{x}))} = \lim_{x \rightarrow 0^+} \frac{x \log(2) - x \log(3)}{\log(1 + (2^x - 1 + x^2 \sin(\frac{1}{x})))} =$$

$$\Rightarrow \log(1 + (2^x - 1 + x^2 \sin(\frac{1}{x}))) \sim 2^x - 1 + x^2 \sin(\frac{1}{x})$$

$$= \lim_{x \rightarrow 0^+} \frac{x \log(2) - x \log(3)}{2^x - 1 + x^2 \sin(\frac{1}{x})} = \lim_{x \rightarrow 0^+} \frac{x \log(2) - x \log(3)}{\frac{2^x-1}{x} \cdot x + x^2 \sin(\frac{1}{x})} = \lim_{x \rightarrow 0^+} \frac{x(\log(2) - \log(3))}{x(\log 2 + x \sin(\frac{1}{x}))} =$$

$$= \frac{\log(2) - \log(3)}{\log(2)}$$